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MODELING AND CONTROL OF A 1-DOF HELICOPTER MODEL USING LABVIEW

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Report on
**MODELLING AND CONTROL OF A 1-DOF HELICOPTER
MODEL USING LABVIEW**



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Abstract

This report is on the development of a model and closed loop control for a Vertical Take-off and Landing (VTOL) system. The closed loop configuration was achieved using a feedback signal from a potentiometric sensor. The sensor has been calibrated before implementing in real time system. System Identification toolkit of LabVIEW was used to estimate the Transfer Function for the system. This estimated transfer function was then simulated using the control design and simulation module of LabVIEW. Closed loop control was affected using a PID controller on the simulation model and it was evaluated on the real time model.

Keywords: VTOL, Closed loop system, Calibration, system identification, control design& simulation, Real-time testing.

1. Introduction

Due to their inherent complexity, aircraft systems are usually broken down into several subsystems to make it easier to analyse. These subsystems can be dealt with individually and then integrated to provide an overall solution. The objective of this project is to develop an experimental setup for modelling, analysis and control of vertical take-off and landing (VTOL) and hover characteristics of aircrafts such as helicopters, quad-rotor and tilt-rotor aircrafts.

The proposed system consists of an electric motor plus propeller combination mounted at one end of an arm. The arm assembly pivots about a rotary potentiometer shaft. The vertical displacement of the rotor is calculated from the pitch angle of the pivot, and is a function of the thrust generated by the rotor, which in turn is a function of the angular velocity of the rotor. Some Real life applications of VTOL are given below:

- Helicopters
- Rockets
- Tiltrotors

1.1. Objective

1. Development of 1DOF VOTOL experimental setup
2. Calibration of Sensor (Potentiometer)
3. System identification of the VTOL Model (finding transfer function etc.)
4. Simulation and control of VTOL model in LabVIEW
5. Real-time study of VTOL model

2. Modelling of VTOL

The equations of motion of the VTOL can be derived from first-principles. A free body diagram of the 1-DOF VTOL device is shown in Fig.1. When the system is in equilibrium we can sum the torques around the pivot point to obtain

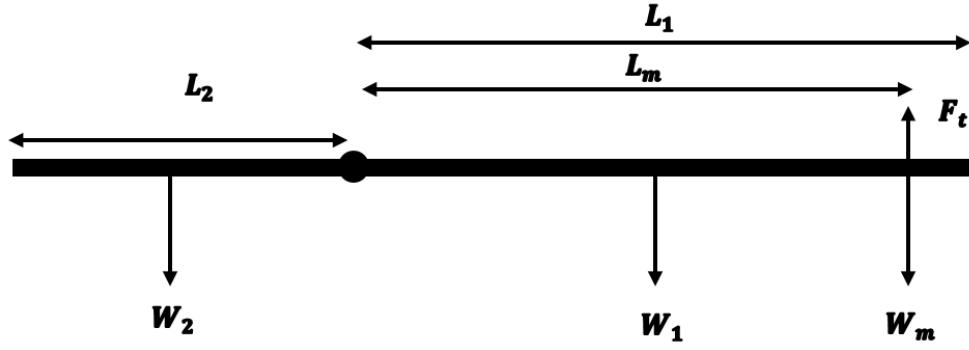


Figure 1-Figure shows the forces acting on VTOL

$$F_t L_m + \frac{1}{2} W_2 L_2 \cos \theta - \frac{1}{2} W_1 L_1 \cos \theta - \frac{1}{2} W_m L_m \cos \theta = 0 \quad (1)$$

$$\tau_t = F_t L_m \quad (2)$$

$$\tau_t + \frac{1}{2} W_2 L_2 \cos \theta - \frac{1}{2} W_1 L_1 \cos \theta - \frac{1}{2} W_m L_m \cos \theta = 0 \quad (3)$$

When the VTOL system is in equilibrium and level to the ground $\theta = 0, \cos \theta = 1$, the torques acting on the system are described as:

$$\tau_{t_hor} + \frac{1}{2} W_2 L_2 - \frac{1}{2} W_1 L_1 - \frac{1}{2} W_m L_m = 0 \quad (4)$$

Where, V_{m_hor} denotes the voltage required to maintain the system in level equilibrium.

For modelling the motor, the transfer function for armature controlled motor (fig.2) was chosen, which gives the relationship between the armature voltage V_m and the angular velocity of the shaft ω .

$$G(S) \equiv \frac{\omega(s)}{V_m(s)} = \frac{K_m}{[(R_a + L_a s)(Js + B) + K_b K_m]} \quad (5)$$

Where, J is the moment of inertia acting about the pitch axis, B is the viscous damping.

Comparing the above transfer function with the general II order system transfer function, we get,

$$G(S) \equiv \frac{\omega(s)}{V_m(s)} = \frac{K_m}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (6)$$

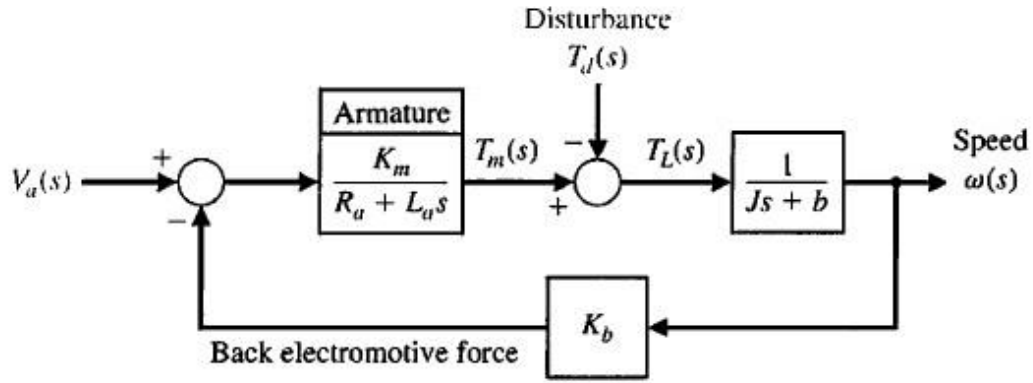


Figure 2-Transfer function of an armature controlled motor [2]

3. Experimental Setup Design

The VTOL experimental setup is designed using CATIA V5 R19. After designing 3D model of experimental setup, acrylic parts of the setup have been made by CNC Machining process. Fig.3 shows the assembled experimental setup. Fig.4 shows the manufactured experimental setup.

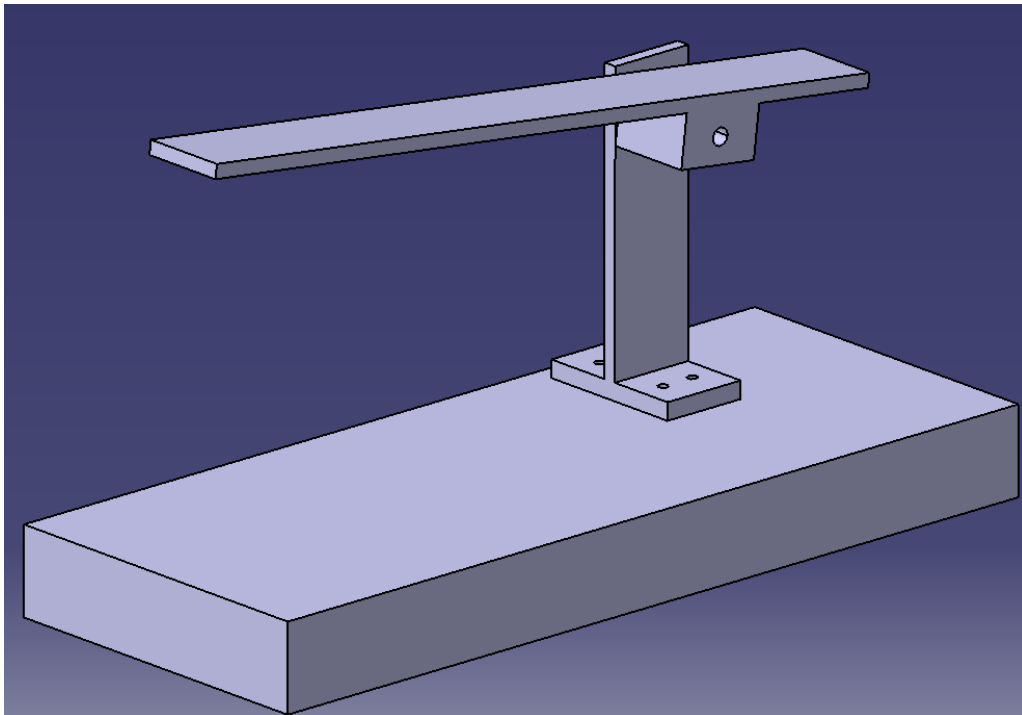


Figure 3-3D Catia assembly drawing of the Experimental setup

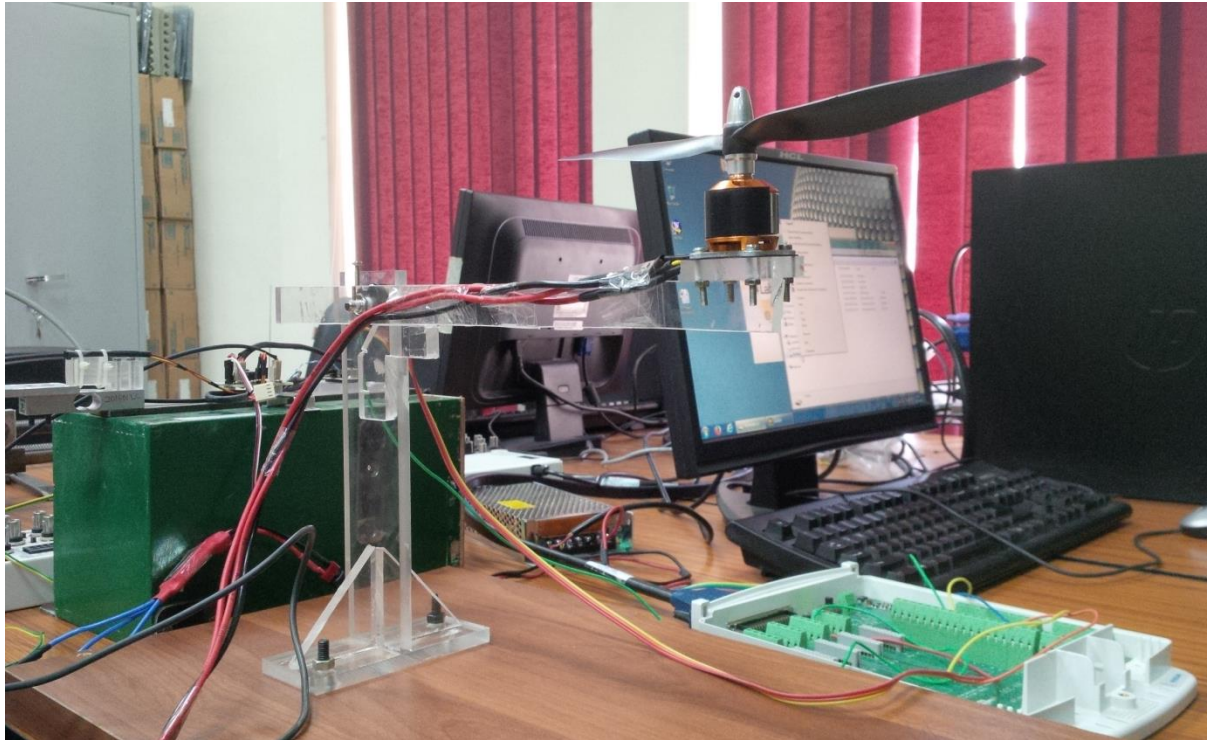


Figure 4-Experimental setup

Main parts of the experimental setup are:

3.1. BLDC Motor

BLDC Motor has been used for the present application. It has a hardened steel shaft, dual ball bearings, and gold plated spring male connectors. The BLDC motor used in the experiments has following specifications.

- Input voltage: 7.4-11.1V DC or 2-3s li-Po Battery
- Max Efficiency Current: 4-10A (>75%)
- No Load Current(10V): 0.5A, MAX CURRENT: 12A/60sec
- Max Watts: 150W
- Internal Resistance: 90m Ω
- Minimum Recommended ESC/Controller: 18A
- Recommended Model Weight: 300-800g

3.2. Propeller

The Propeller used in the experiments has following specifications.

- Item Weight: 41 grams
- Length: 10 inch (25.4cm)
- Slope: 4.5 inch (11.43cm)

3.3. Electronic Speed Controller (ESC)

Electronic Speed Controller: Most of the ESCs need a 50Hz frequency i.e. a 20ms time period cycle and the speed depends upon the duty cycle we provide. 1ms (5% duty cycle) will

reduce its speed to minimum or even stop(depends on the ESC model) while a 2ms (10% duty cycle) pulse will run the motor on its full speed. Values between them give the variation of the speed. The ESC used in the experiments has following specifications.

- Power Supply: 7.4-11.1V DC or 2-3s li-Po Battery
- Drive Current: 30A (Max: 40A/10sec)
- Battery Eliminator Circuit(BEC): 5A/3A

3.4. Potentiometer

The Potentiometer (B502-B5K-20F) used in the experiments have following specifications.

- Nominal Resistance Range: 5K
- Resistance Deviation: $\pm 5\%$
- Rated Power: 2W
- Laps: 1 Ring
- Electrical angle: $280^\circ \pm 10^\circ$
- Mechanical angle: $300^\circ \pm 5^\circ$

4. Calibration of the Sensor

Before using the potentiometer, it has been calibrated. Potentiometer is deflected at some known angles and the corresponding voltages have been recorded into a text file using **Potentiometer_Calib.vi**. The text file has been read by using **Analysis_Calib_Data.vi**. In this VI **Linear Fit.vi** and the **correlation coefficient.vi** has been used for the data fitting and Regression analysis. Fig shows the front panel and block diagrams of the Vis used in the calibration.

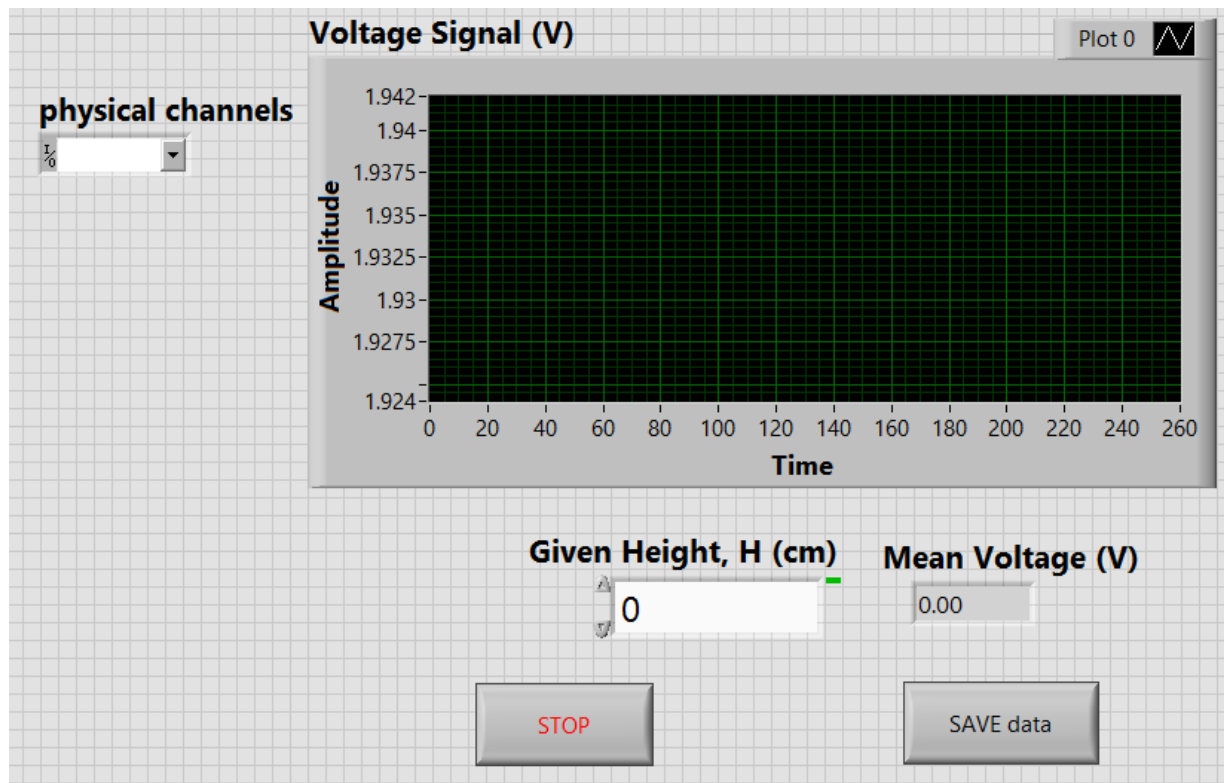


Figure 5-Front Panel of Potentiometer_Calib.vi

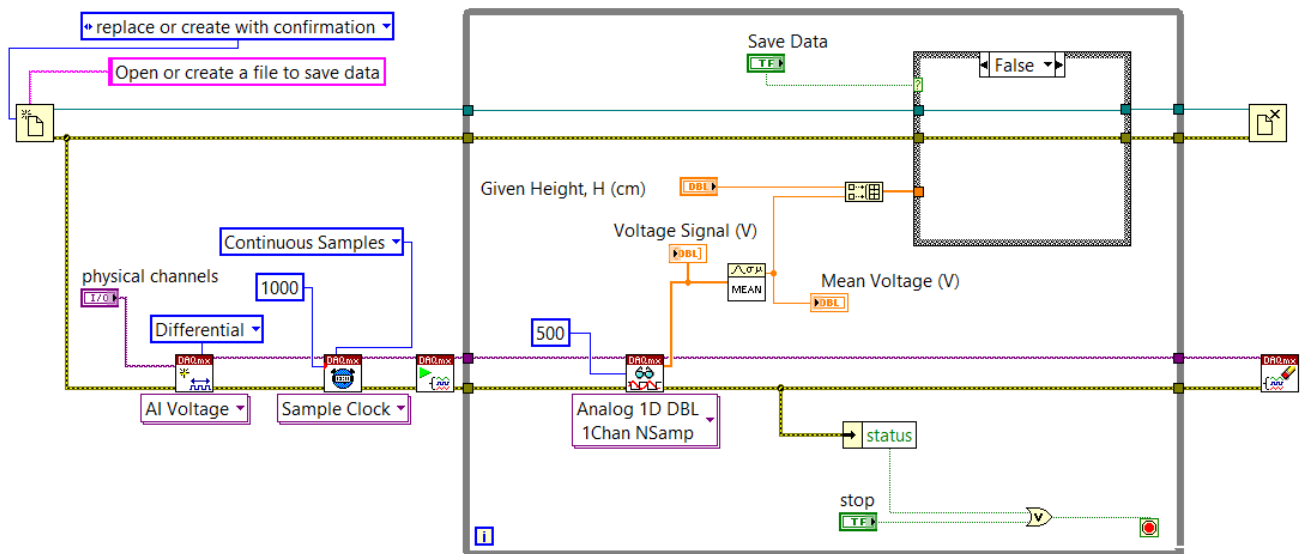


Figure 6- Block diagram of Potentiometer_Calib.vi

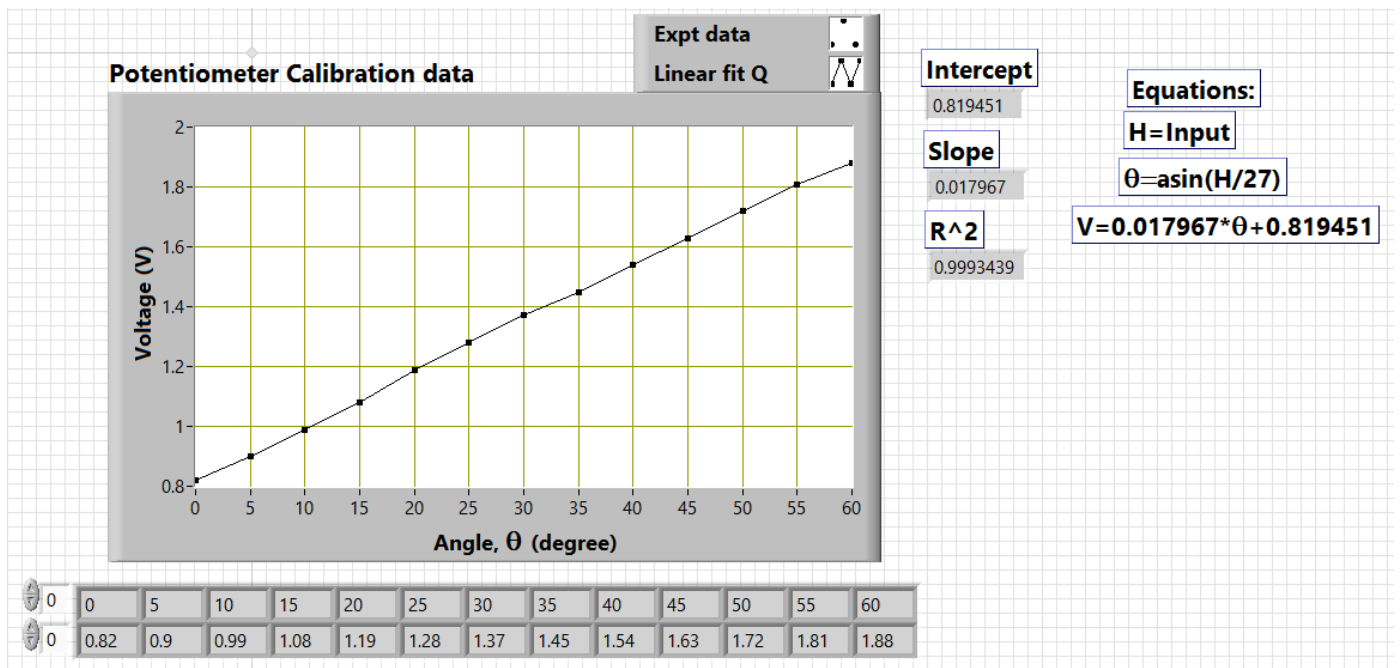


Figure 7- Front panel of Analysis_Calib_Data.vi

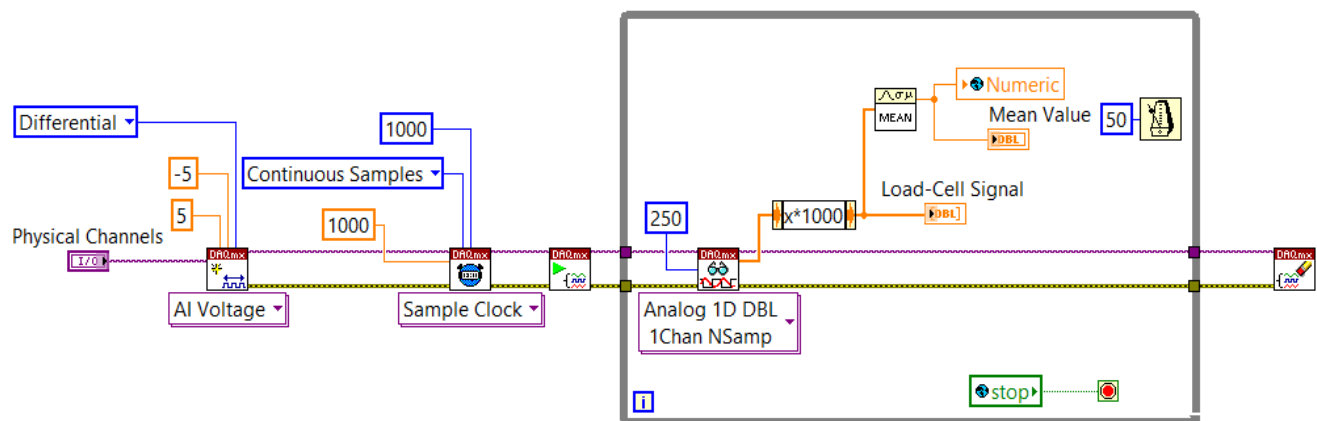


Figure 10- Block diagram of Loadcell_Test.vi

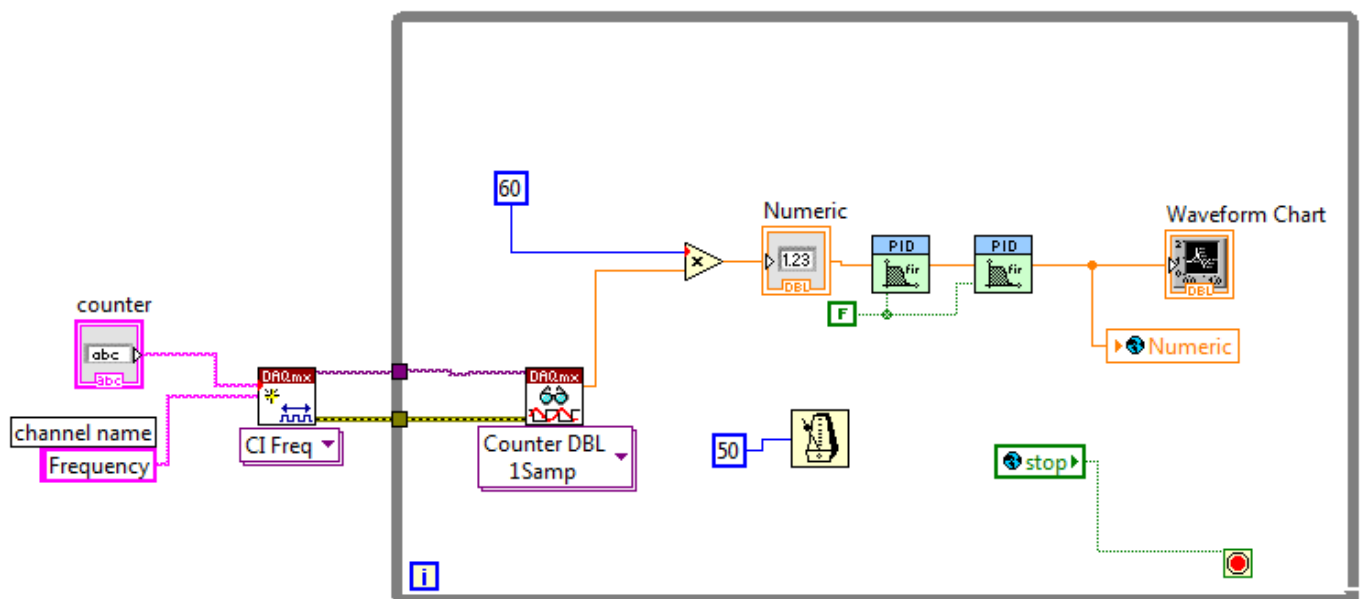


Figure 11- Block diagram of RPM Counter.vi

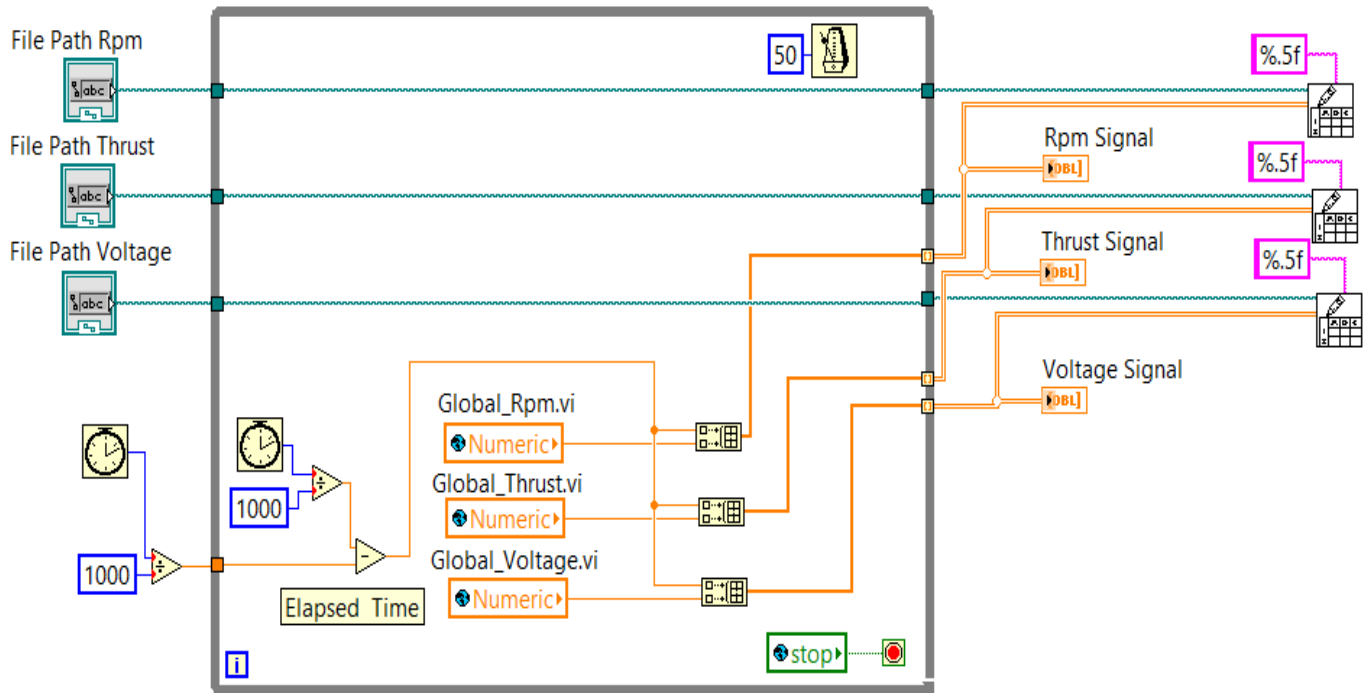


Figure 12- Block diagram of Data Acquisition.vi

5.2. Identification of Transfer Function

The transfer function of the model has been calculated using *transfer_function.vi* and acquired data.

$$G(S) \equiv \frac{\omega(s)}{V_m(s)} = \frac{K_m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$K_m = 1922.88$$

$$\zeta = 0.671$$

$$\omega_n = 4.5 \text{ rad/s}$$

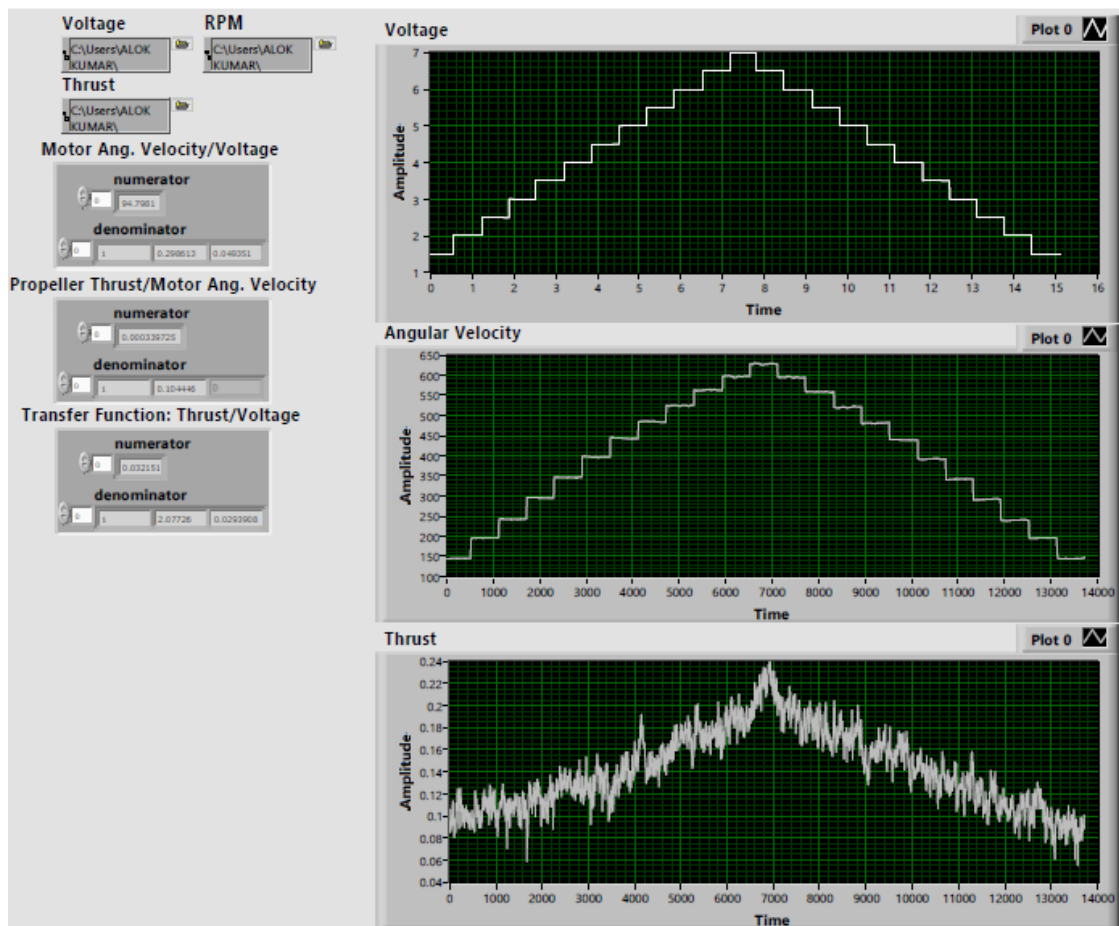


Figure 13- Front panel of `transfer_function.vi`

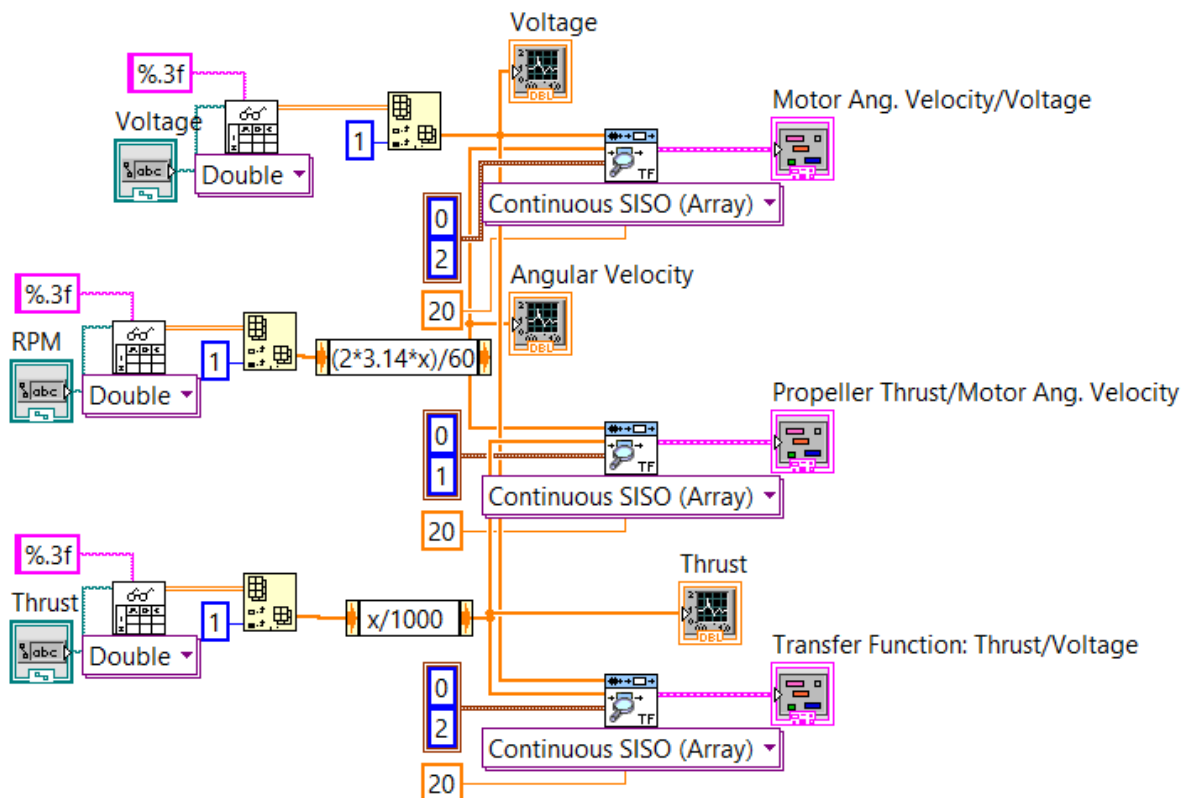


Figure 14- Block diagram of `transfer_function.vi`

5.3. Control Design and Simulation

After getting the transfer function of the model system simulation work has been done by using ‘*Control Design & Simulation*’ module of LabVIEW. *PID_Simulation.vi* has been used for the simulation. Fig.15 and Fig.16 shows the front panel and block diagram of the *PID_Simulation.vi* respectively. Waveform graph in fig.14 shows the variation of response for given set point.

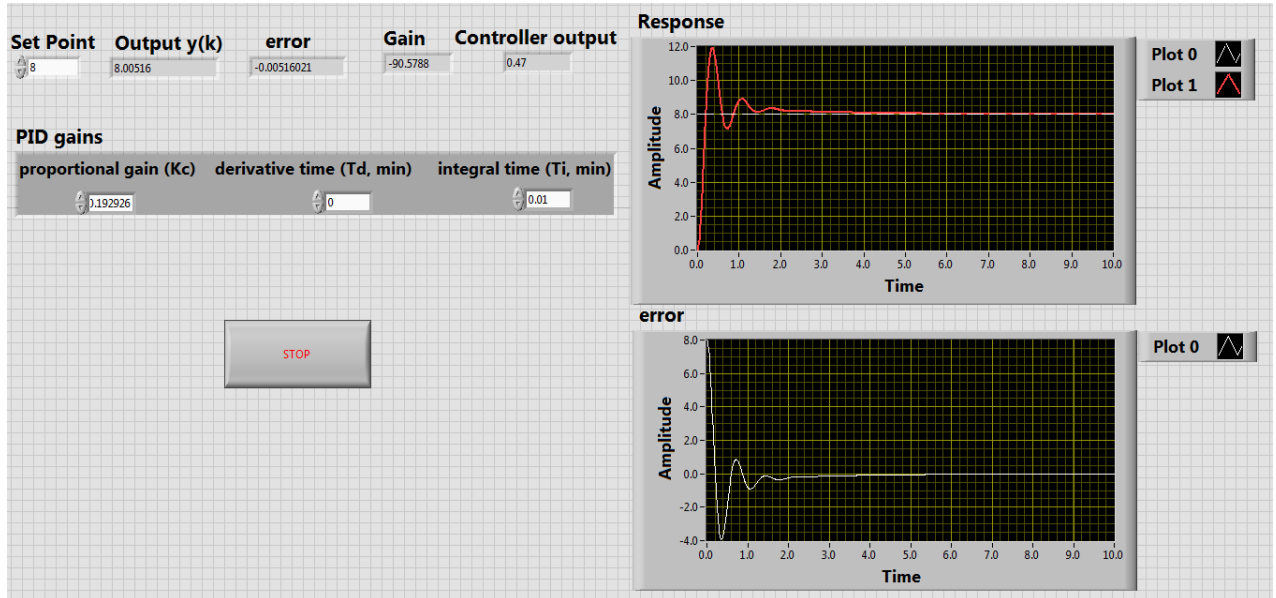


Figure 15-Front panel of *PID_Simulation.vi*

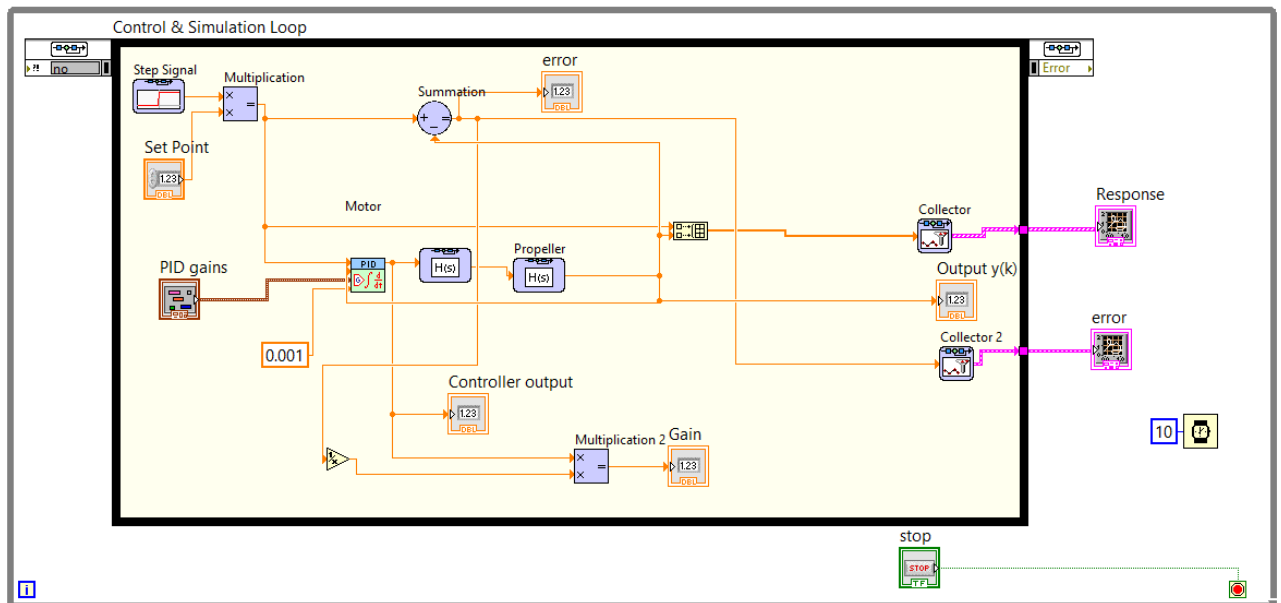


Figure 16-Block diagram of *PID_Simulation.vi*

6. Real-time study

For implementation and control of the VTOL model, the *BLDC_Control.vi* was built and using a *PID* controller and appropriate gains, the desired system response in terms of pitch angle was achieved in real time. Fig.17 and Fig.18 shows the front panel and block diagram of

the *BLDC_Control.vi* respectively. Waveform graph in fig.16 shows the variation of response for given set point.

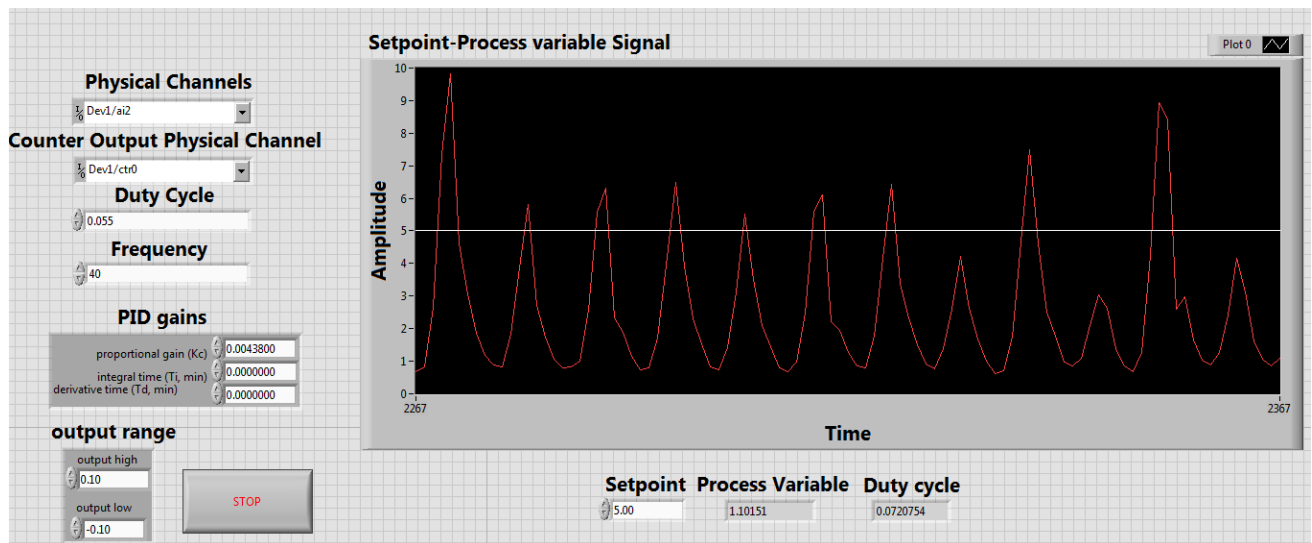


Figure 17-Front panel of *BLDC_Control.vi*

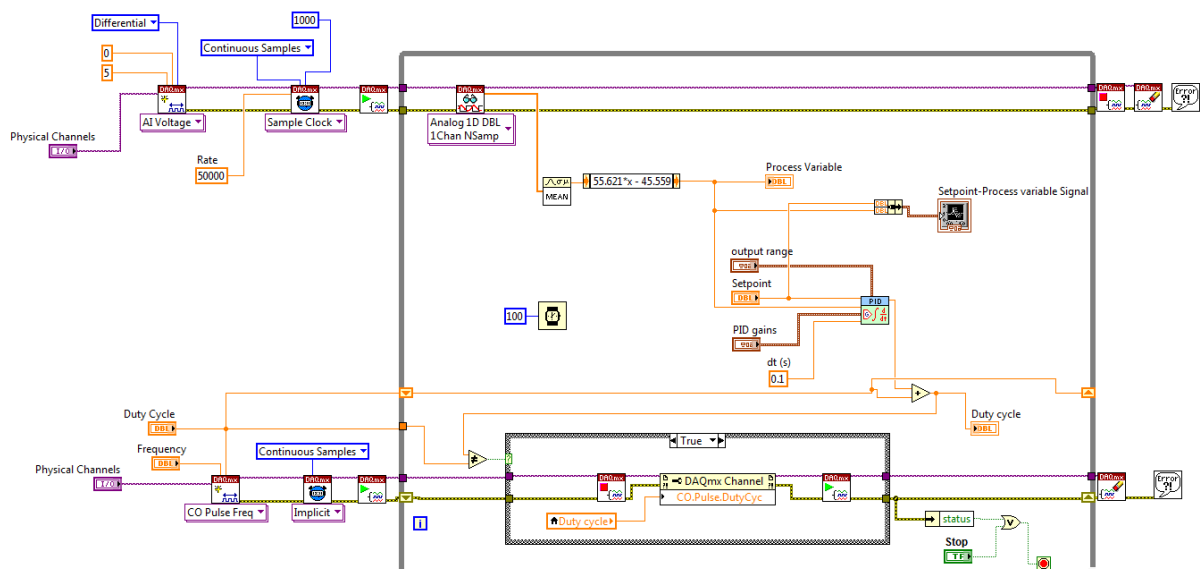


Figure 18- Block diagram of *BLDC_Control.vi*

7. References

1. Abara, Daniel Nsor, "The development of a model and controller of a vertical take-off and landing (vtol) system Using LabVIEW and Simulink", The University of Manchester, 2015.
2. R. Dorf, R. Bishop, "Morden Control Systems", 12th Edition, Prentice Hall, 2011.
3. QNET Vertical take-off and landing trainer for NI-ELVIS, Quanser Inc., 2009.

Appendix

Table-1: Values of different parameters

Parameters	Value
W_1	50.976 gm
W_2	13.97 gm
W_m	100 gm
L_1	27 cm
L_2	7.4 cm
L_m	25.25 cm